The Future of Renewable Energy

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Solar and wind energy capacity has surged globally, with installed capacity reaching 2.5 terawatts in 2023, a 15% increase from the previous year. Investment in renewable energy exceeded $500 billion for the first time, driven by government incentives and falling technology costs. The levelized cost of electricity (LCOE) for utility-scale solar has dropped to $0.033/kWh, making it the cheapest electricity source in history.

Battery storage technology has kept pace with renewable expansion. Global battery storage capacity reached 85 GWh by end of 2023. Lithium iron phosphate (LFP) batteries now dominate the market at 62% share, valued for safety and long cycle life exceeding 4,000 cycles. Grid-scale projects increasingly combine solar/wind with storage to provide firm, dispatchable renewable power around the clock.

Global Renewable Energy Statistics 2024

Energy Source	Capacity (GW)	Growth YoY	LCOE ($/kWh)	Jobs Created
Solar PV	1,419	+28%	$0.033	4.2M
Wind (Onshore)	892	+12%	$0.041	1.4M
Wind (Offshore)	72	+24%	$0.081	0.3M
Hydropower	1,392	+2%	$0.048	2.0M
Geothermal	15	+4%	$0.071	0.1M
Total Renewable	**3,790**	**+16%**	**-**	**7.9M**